

Whence the Matrix?

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DELPH-IN Fefor Meeting
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Overview

- [Three roles for the Matrix

- [Changes awaiting release

- [Planned future work

Three Roles for the Matrix

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- [Point of contact for 'big' grammars
- [Starter kit for new grammars
- [Research on computational typology

Point of Contact: Benefits

- [Role of Matrix in standardizing semantic representations
- [Feedback from mature grammars to Matrix
- [Potentially: Auto-generation of certain kinds of transfer rules

Point of Contact: Wishlist

- [Regular Matrix releases

- [Discussion of proposed changes on matrix@delph-in.net

- [Software support for synching to new releases

- e.g., tracking of changes, including constraints moved out from matrix.tdl to other files

Starter Kit: Benefits

- [Quick start-up

- [Consistency with mature grammars

- [Multilingual grammar engineering courses

- [... feedback to the Matrix from many, typologically diverse languages

Starter Kit: Wishlist

— [More phenomena (Matrix & 'Modules')

— [Fewer bugs

— [Documentation

Computational Typology

— [Choose phenomena (word order, negation, pro-drop, coordination, ...)

— [Research expressions of the phenomena across the world's languages

— [Implement variations

— [Develop customization questionnaire:

<http://www.delph-in.net/matrix/modules.html>

Hypothesis Testing

- [Can we implement all known variants, given the constraints of tdl and the Matrix core?
- [Are all variants for all phenomena consistent with each other?
- [Do inconsistent combinations correctly predict impossible language types?
- [Are the implementations consistent with actual languages?

Changes Awaiting Release

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- [Lexical rules

- [Realizing complements out of order

- [Determinerless NPs

- [Messages

- ['Modules'

Lexical Rules

- [Want effect of 'copy up everything except what I explicitly change'
- [Hierarchy of types for copying particular features
- [Need documentation or tdl-generating software to handle the addition of further features by users

Lexical Rules: Examples

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```
same-cont-lex-rule := lex-rule &  
  [ SYNSEM.LOCAL.CONT [ HOOK #hook, MSG #msg ],  
    DTR.SYNSEM.LOCAL.CONT [ HOOK #hook, MSG #msg ] ].
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same-local-lex-rule := same-cat-lex-rule &  
                        same-cont-lex-rule &  
                        same-ctxt-lex-rule &  
                        same-agr-lex-rule.
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Lexical Rules: Examples

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same-local-lex-rule := same-cat-lex-rule &  
  same-cont-lex-rule &  
  same-ctxt-lex-rule &  
  same-agr-lex-rule.  
  
no-ccont-lex-rule := lex-rule &  
  [ C-CONT [ RELS <! !>,  
    HCONS <! !> ] ].
```

Lexical Rules: Examples

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```
local-change-only-lex-rule := same-non-local-lex-rule &  
                             no-ccont-lex-rule.
```

```
cat-change-only-lex-rule := local-change-only-lex-rule &  
                             same-ctxt-lex-rule &  
                             same-cont-lex-rule &  
                             same-agr-lex-rule.
```

```
head-change-only-lex-rule := cat-change-only-lex-rule &  
                             same-val-lex-rule &  
                             same-hc-light-lex-rule &  
                             same-posthead-lex-rule.
```

Complement Order

— [Head-comp phrase picks of first COMPS element

— [Allow free order of COMPS by creating head-2nd-comps-phrase which picks of 2nd

— [Eventually: head-3rd-comps-phrase?

— [head-opt-2nd-comp phrase might also be useful

Determinerless NPs

- [basic-bare-np-phrase takes NOM and creates NP with quantifier semantics
- [Specific grammars constrain the quantifier relation supplied as well as syntactic constraints on the rule's application
- [In some cases, multiple bare-np-phrases are warranted

Messages 1

- [Moving towards uniform introduction of messages via non-branching rules

- [Cross-classification of clause types and phrase types

- doesn't work for free word order languages

- is a headache at the cross-linguistic level

Messages 2

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- [Head-subj, head-opt-subj, head-filler all underspecified because of cross-classification in the ERG
- [Add a module (used by most grammars) patching that up?
- [... unless we change the representation of messages all together

'Modules'

- [Non-universal, yet recurring patterns
- [Organized around 'phenomena'
- [Not really that modular (un système où tout se tient)
- [Need a new name

Current 'Modules'

—— [Word order (major constituents, det-N, P-NP)

—— [Coordination

—— [Sentential negation

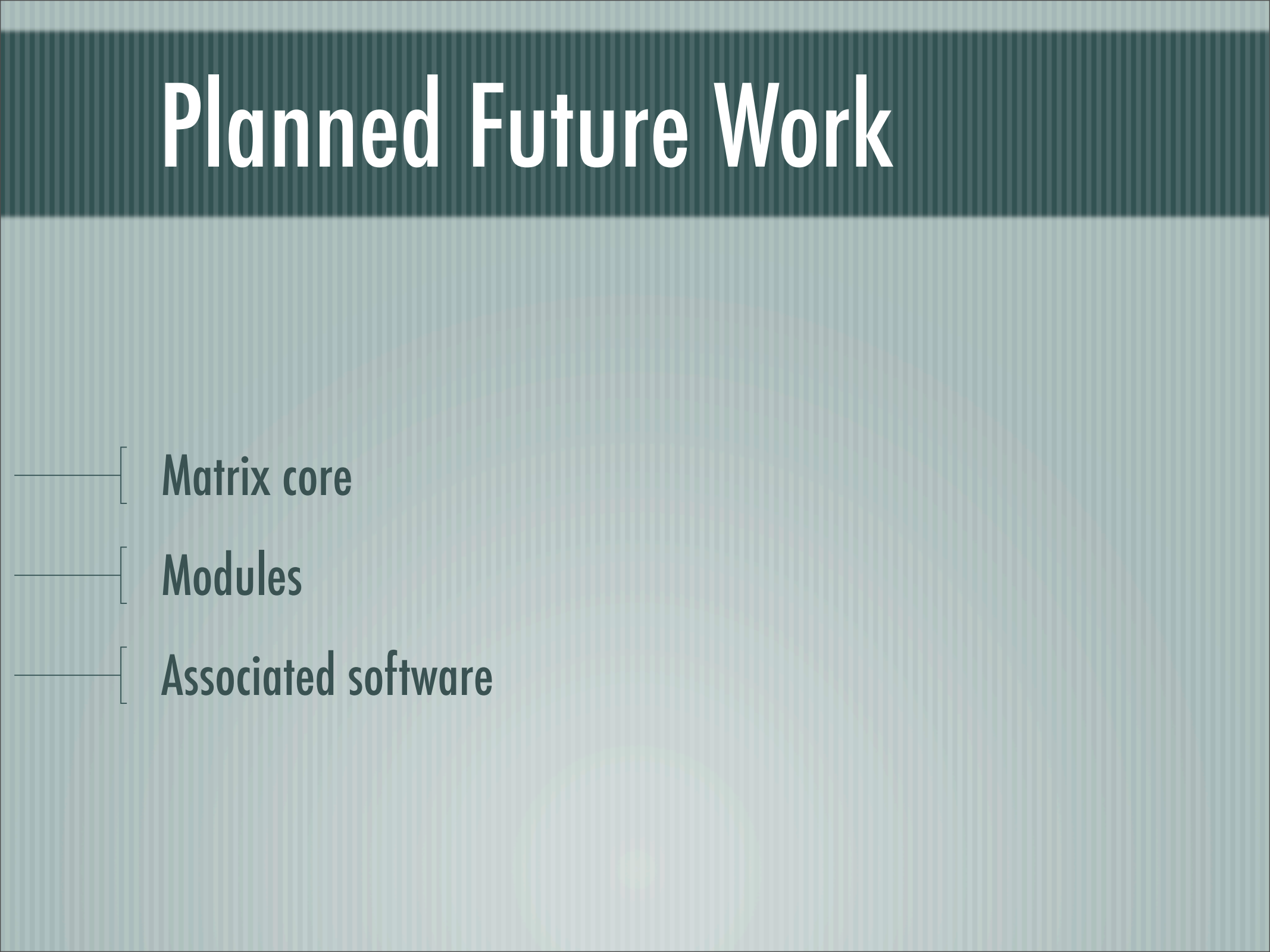
—— [Intransitive & transitive verbs

—— [Determiner 'optionality'

—— [Case-marking adpositions

Planned Future Work

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— [**Matrix core**

— [**Modules**

— [**Associated software**

Matrix core

- [Clean up message introduction

- [Extend lexical rules hierarchy

- [Rework representation of discourse status

Modules

— [This summer: Rework customization script

— [Start building modules test suite resource (cf. Laurie's talk)

— [Near-term modules: Case, pro-drop, clausal semantics, more verb types, modifiers (adjectives, adverbs), pronouns

— [Longer-term modules: Agreement, relative clauses, tense/aspect, compounding, modals, noun incorporation, numeral classifiers... (Scott Drellishak's PhD work)

Associated software

- [Support for updating to new versions of matrix core?
- [Mass lexicon import in connection with the customization script (interface to LexDB?)
- [Autogeneration of transfer rules for 'grammatical' aspects of MRS based on customization choices for two grammars
- [Automatic lexical acquisition for immature grammars

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— [... but is it worth the effort to everyone?